Complex mechanical learning with conservative KAM certification and dissipative magnetic feedback

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Abstract

Connecting a learned mechanical model to stability and feedback requires error bounds for the functions actually deployed. We study this connection for a planar magnetic rod with prescribed configuration-dependent inertia. A constrained complex network first learns a higher-order potential correction, followed by a separate positive-semidefinite Rayleigh decoder. Native COMSOL weak observations yield a validated finite inverse that retains potential nuisance, quadrature consistency and analytic remainders. For the frozen learned–reference pair, the resulting complex-domain derivative bounds enter an a posteriori KAM proof for their conservative Hamiltonian family. Forty rest-start configurations provide a separate observation-only gradient validation that passes this conservative transport. The learned correction reduces gradient RMSE by approximately 5.3×10^3 relative to its fourth-order prior on these observations. A separate known-reference sixth-moment integration certifies a population-risk bound for this same frozen pair. The dissipative model satisfies a mechanical power identity, with identified errors transported to physical magnetic feedback. A local sampled-flow certificate and a numerical energy ledger give conditional continuing practical invariance and entire-hold tracking. The executed evidence comprises 200 validated holds and two further closed-loop continuation holds. The same inverse accepts data-adaptive outputs with individually certified analytic remainders. State-dependent real and complex damping heads improve on a constant baseline, with the real head performing better in this comparison. A separate four-cell conservative comparison favors the complex models under its original budget, whereas a precision-centered prior-assisted comparison favors the real model. Neither repaired generic model improves on the fourth-order baseline. The results concern the specified finite-Gauss reference and declared viscosity; statistical generalization and a general advantage of complex representations remain separate questions.

Keywords

magnetic soft robots, complex Hamiltonian learning, Rayleigh dissipation, a posteriori KAM theory, sampled feedback

1 Introduction

Magnetic actuation enables remote manipulation by applying fields and gradients to magnetized bodies, allowing the source of mechanical power to remain away from the moving instrument. The forces and torques available to a robot depend on both its magnetic properties and the field generated by the actuation system (Abbott et al. 2020). For a flexible body, this interaction is coupled to deformation, inertia and energy loss. Ferromagnetic soft continuum robots demonstrate how distributed magnetization can support steering and navigation in constrained environments (Kim et al. 2019). Predictive models are consequently useful for relating an applied magnetic command to the evolving mechanical state. A learned model intended for feedback must also provide an account of how its remaining error affects the conclusions drawn from that model.

Rod elasticity and finite-strain beam dynamics provide a mechanical starting point for describing flexible bodies (Antman 2005; Simo 1985). Reduction to a small set of coordinates makes identification and feedback more tractable, while retaining dependencies inherited from the deformation geometry. In particular, the kinetic energy can depend on configuration through an inertia matrix. This dependency changes the canonical momentum equation and the conversion between measured velocity and momentum. We study a planar reduced magnetic-rod chart with a prescribed positive mechanical metric. Its known inertia and elastic prior define the baseline model, and learning supplies a higher-order potential correction and a separate

dissipation factor. The finite reduction provides the common object for observation, training, mathematical verification and feedback.

Hamiltonian neural networks offer a useful approach to this task by learning a scalar energy and differentiating it to obtain a canonical vector field (Greydanus et al. 2019). The energy representation links the components of the predicted motion and gives the unforced conservative model an exact differential power identity. For the present system, this representation also permits the known configuration-dependent kinetic term to be retained while the uncertain potential is learned. Keeping that mechanical information explicit reduces the part of the function that must be inferred. It also specifies the coordinates, derivatives and energy gauge to which an eventual error certificate applies. Conservation of the learned scalar is a structural property; closeness to the reference mechanical system requires additional information from observations.

Dissipation and magnetic input extend the identification problem. Dissipative SymODEN incorporates energy loss and control within a port-Hamiltonian computation graph, and port-Hamiltonian neural networks separate a stationary energy from dissipative and driving terms (Zhong et al. 2020; Desai et al. 2021). These developments motivate a model in which the conservative energy, Rayleigh dissipation and magnetic port retain their distinct mechanical roles. We use a configuration-dependent positive-semidefinite decoder for the dissipation matrix. The associated Rayleigh function is quadratic in velocity, so its velocity derivative gives a signed dissipative force. The energy is trained first and held fixed during the subsequent dissipation fit. This sequential procedure provides a definite deployed pair whose power relation can be checked and whose two sources of approximation error can be transported into the control analysis.

Complex-valued networks supply the trainable features in both stages. The complex affine operations and real-coordinate implementation follow the building-block perspective of Trabelsi et al. (2018). Here the resulting potential is real valued on physical states, and a positive decoder converts the dissipation output into a real mechanical matrix. Literal complex hyperbolic tangent activations also make pole exclusion and derivative control part of the analysis. These properties depend on the actual weights and verification domain. The role of the complex representation is therefore assessed through its specified architecture, fitted output and comparisons, while the canonical and power identities follow from the mechanical construction.

Optimization theory helps clarify the connection between that representation and its training record. The complex-gradient analysis of Zhang et al. (2022) motivates a convention-explicit descent component with a smooth objective and a strict decrease margin. The certified energy and dissipation fits use recorded Armijo updates, including damped Gauss–Newton steps for the dissipation head. The generic full-phase factorial comparators use a separate fixed-budget Adam protocol. Their accepted finite outputs are the inputs to verification. This formulation lets the mathematical error analysis apply to the functions actually returned by training, including a nonzero residual, without requiring an unverified claim of global optimization or asymptotic stationarity.

There are also two distinct uses of complex coordinates in the mechanical analysis. Expressing real canonical position and momentum as a complex state yields a Wirtinger equation for the real energy. Analytic KAM estimates instead extend the real canonical variables to independent complex arguments. We derive both interpretations for the same nonseparable kinetic energy and specify the real-output continuation used by the networks. This distinction preserves the configuration derivative of inertia and identifies the analytic domain on which the later derivative and torus estimates are valid.

Time discretization introduces a further structural question. A Hamiltonian vector field can be integrated by a method whose discrete map preserves the symplectic form; implicit midpoint is a classical example (Hairer 2010). We construct that scheme for the present configuration-dependent inertia and derive local solvability, symplecticity and a second-order global error estimate on an explicit common domain. A finite nonlinear-solver residual contributes its own error allowance. This gives a precise conservative numerical component alongside the continuous model. The observation solvers and the continuous-flow KAM verification retain their separately stated contracts, so the mathematical role of each approximation remains visible.

The principal difficulty is to carry learning accuracy into the norms needed by a stability argument. Chen et al. (2022) connect nonzero-loss Hamiltonian learning to KAM theory through approximation, statistical risk control and regularity. Their route makes the sampling law, hypothesis-class bounds and derivative regularity essential parts of the conclusion. In a trajectory experiment, however, successive windows share the same motion and prescribed initial conditions. The resulting empirical objective directly constrains observed balances, whereas KAM requires analytic vector-field and embedding-defect estimates on a whole domain. Resolving

this difference calls for quantitative identification and remainder bounds as well as a training loss.

We first develop such a connection for the available weak observations. Integral momentum balances use recorded configuration, observed velocity, prescribed input and time. They avoid acceleration labels and isolate the finite potential and symmetric matrix-polynomial coordinates that influence the measured balances. A validated scaled left inverse then separates those contributions. The derivation retains quadrature consistency, observation errors and analytic remainders. It also keeps the finite potential nuisance used during the dissipation fit: recovering the deployed Hamiltonian requires its saved coefficient to be subtracted explicitly. The resulting error bound therefore concerns the actual learned energy rather than an energy silently modified by profiling.

A complementary rest-start observation design supplies conservative gradient information. At zero initial momentum and zero input, the initial momentum derivative equals the negative potential gradient. Observed short-window configurations and velocities determine a forward label, with the actual recorded clock and an explicit truncation allowance. Retaining both gradient components yields a rectangular finite-output design whose left inverse is validated for the realized configurations. Nonzero signed residuals, label errors, observation-point tails and complex-domain remainders then give a checkable bound on the whole candidate function. The same algebra applies simultaneously to data-adaptive outputs that each obtain valid remainder certificates. This is a useful output-acceptance statement even when a common bounded-loss description of every raw network parameter is unavailable. Random excitation is established for this finite representative under the declared discrete sampling law. Stability is then checked using the realized inverse, keeping rank probability and numerical amplification as separate design quantities.

For the conservative endpoint, classical KAM theory describes the persistence of suitable quasiperiodic invariant tori under small Hamiltonian perturbations (Pöschel 2001). An a posteriori formulation starts from an approximate embedding and validates its residual, arithmetic and geometric hypotheses (Haro and Luque 2019). We transport the identified potential error through ordered derivative bounds for the same learned–reference family. The calculation retains the complete embedding invariance defect, including its Fourier and parameter tails, together with torsion and domain margins. In this way, observations contribute a quantitative function-error component, while a separately verified embedding supplies the torus witness. Each new learned output requires those model-dependent conditions to remain consistent with its own Hamiltonian.

The forced dissipative endpoint is governed by the power relation. Positive-semidefinite dissipation gives the learned system a nonpositive unforced energy rate, and the certified potential and dissipation errors provide allowances for the reference dynamics. A magnetic work constraint then limits the energy supplied during a held command. The feedback argument combines this constraint with local contraction and bounded-error estimates through an induction on the state and a carried energy ledger. Observation boxes, admissible inputs and positive hold-clock bounds enter throughout each interval. The resulting practical-invariance statement has a continuous-time meaning even though decisions are made at discrete boundaries. Software return and arithmetic certificates describe the numerical decision interface; sensing, computation and actuation latency remain outside the stated hold model.

The numerical study uses COMSOL Global Equations for the specified reduced chart. The potential and dissipation fits use distinct trajectory cohorts, and reserved trajectories are evaluated after fitting. Native records support independent whole-interval validation of the relevant trajectories and held commands. The rest-start experiment retains every sampled configuration and seals its observation-only labels before evaluating known-reference discrepancies. A higher-order-potential ablation examines the value of the learned correction relative to the prescribed elastic prior; comparisons among constant, real and complex dissipation heads examine their declared parameterizations. A separate real/complex by quadratic-prior factorial study compares generic full-phase models under a common training budget. Together these tests connect fitted functions, observation requirements and mathematical acceptance rather than treating trajectory agreement as the complete validation target.

The contribution is a mechanically consistent chain from sequential complex energy–dissipation learning to quantitative conservative verification and magnetic feedback. It combines a nonseparable reduced Hamiltonian, a structure-preserving numerical component, observation-to-function acceptance with analytic tails, nonzero-error KAM validation and a dissipative sampled-control argument. Conditional statistical learning statements identify which additional sampling and class conditions would support population-risk inference, while the executed results use their actual deterministic certificates. The conclusions apply to the stated finite-Gauss mechanical reduction and numerical viscosity assumption. Continuum accuracy,

physical parameter calibration and real-time hardware implementation define the next modeling and experimental steps.

2 Complex Hamiltonian construction and symplectic discretization

2.1 Reduced rod mechanics and physical normalization

The physical reference is a planar, inextensible and unshearable Kirchhoff rod with a fixed base and a magnetic tip moment. Two prescribed shape functions supply the dimensionless coordinates $q \in \mathbb{R}^2$. Writing $\beta_i(s) = f_i'(s)$, its angular and centreline descriptions are

$$\theta(s,q) = \beta(s)^{\mathsf{T}} q, \qquad r(s,q) = \int_0^s \begin{pmatrix} \sin\theta(\xi,q) \\ \cos\theta(\xi,q) \end{pmatrix} \mathrm{d}\xi. \tag{1}$$

The spatial components are transverse x and axial z. The trigonometric tangent retains finite rotations, while the prescribed two-coordinate chart restricts the admissible shapes. No axial stretching or independent shear variable is added. The construction follows rod kinematics Antman (2005); Simo (1985); all conclusions below concern its specified finite reduction.

The cross-section follows the local tangent, so its angular velocity is linear in the coordinate velocities even when the centreline undergoes finite rotation. This accounts for the constant rotary contribution below. The bending constitutive law remains quadratic in curvature within this chart. Gravity, field gradients and additional body-force terms are absent from the reference equations. The free end receives the magnetic moment; its rotation is not prescribed as an essential boundary condition.

2.1.1 The fixed quadrature and inertia The shape functions are piecewise cubic on 32 cells of the 0.08 m rod. At a material location, the configuration Jacobian is

$$J(s,q) = \partial_q r(s,q) = \int_0^s \begin{pmatrix} \cos\theta(\xi,q) \\ -\sin\theta(\xi,q) \end{pmatrix} \beta(\xi)^{\mathsf{T}} \mathrm{d}\xi. \tag{2}$$

The implemented $J_{ek}(q)$ uses eight-point Gauss quadrature: completed cells contribute their full outer sums, and the current cell contributes a nested eight-point sum up to its outer node. With cell length ℓ_e and positive stored weights w_k, define

$$\begin{aligned} D(q) &= \sum_{e,k} \ell_e w_k J_{ek}(q)^{\mathsf{T}} J_{ek}(q), \\ G(q) &= f_m D(q) + K_r = S(q) S(q)^{\mathsf{T}}. \end{aligned} \tag{3}$$

Here K_r is the constant normalized rotary-inertia matrix, and $f_m > 0$ is the stored translational factor, nominally $\omega_0^2 \rho A_c / E_0$. The notation S always denotes the lower Cholesky factor, not a further modal scaling. The known G is positive definite on the certified configuration box $|q_i| \le 1/10$, and S has strictly positive diagonal. The Gauss nodes, weights and mechanical coefficients are frozen binary64 values. Their exact stored values define the analytic reference; freshly evaluating a continuum integral or a parameter product would define a different reference. In particular, the stored f_m is retained in all identities involving $G - K_r$.

Translational kinetic energy is assembled from the velocities of material points, using their configuration Jacobians before the Gram products are formed. Consequently its inertia depends on configuration even though the two shape functions themselves are fixed. The same metric is used when constructing native trajectories, converting observed velocity to momentum, evaluating the learned Hamiltonian and verifying its conservative flow. The required derivatives differentiate this configuration dependence.

2.1.2 Canonical energy and dimensional quantities Let t denote SI time and set $\tau = \omega_0 t$, with $\omega_0 \simeq 11.417287282852458\,\mathrm{s}^{-1}$ and $E_0 = \mu B_0 \simeq 6.0 \times 10^{-7}\,\mathrm{J}$. Here μ is the tip magnetic dipole magnitude and B_0 the constant bias field. These scales and the shape functions are fixed before learning. A prime denotes $\mathrm{d}/\mathrm{d}\tau$, so $v = q' = \dot{q}/\omega_0$. Dividing physical energy by E_0 gives

$$\begin{aligned} p &= G(q) v, \\ H_*(q,p) &= \tfrac{1}{2} p^{\mathsf{T}} G(q)^{-1} p + V_*(q), \\ V_*(q) &= \tfrac{1}{2} q^{\mathsf{T}} B q + 1 - \cos(d^{\mathsf{T}} q). \end{aligned} \tag{4}$$

The fixed $B \succ 0$ is the normalized bending matrix, and the fixed vector d maps the chosen coordinates to tip rotation. Both are the exported coefficients of this chart. Thus the full bias-field cosine and the configuration-dependent inertia are retained. The momentum p is dimensionless canonical momentum, not SI velocity; if P_{SI} is the momentum conjugate to the same dimensionless q, then $p = (\omega_0/E_0) P_{\mathrm{SI}}$.

Equivalently, the dimensionless Lagrangian is $L_*(q,v) = v^{\mathsf{T}} G(q) v/2 - V_*(q)$, and the momentum definition in (4)

is its Legendre transform. In particular, configuration derivatives of the kinetic energy contribute to the momentum equation. They cannot be replaced by the force of a constant-mass oscillator. Prescribing the inertia also makes the learning problem specific: a potential correction and a dissipative decoder are identified relative to known mechanics, rather than identifying an arbitrary mass matrix from trajectories.

Distributed viscous drag supplies the physical Rayleigh coefficient $C_{\mathrm{phys}} = c_{\mathrm{line}} D(q)$, whose units are $\mathrm{J\,s}$: J has units m, D has units m^3, and c_{line} has units $\mathrm{N\,s/m^2}$. Its normalized damping is

$$\begin{aligned} R_*(q) &= \frac{\omega_0 c_{\mathrm{line}}}{E_0} D(q) \\ &= a\{G(q) - K_r\}, \\ a &= \frac{\omega_0 c_{\mathrm{line}}}{E_0 f_m}, \\ \mathcal{D}_*(q,v) &= \tfrac{1}{2} v^{\mathsf{T}} R_*(q) v. \end{aligned} \tag{5}$$

The declared $c_{\mathrm{line}} = 0.0006911873093836746\,\mathrm{N\,s/m^2}$ is a numerical-study assumption inherited from a nominal small-slope damping estimate, not a measured material calibration. The present reference sets tip drag $c_{\mathrm{tip}} = 0$ and additional mass-proportional drag $\gamma_{\mathrm{add}} = 0$. Its damping is the matrix in (5); no constant scalar-drag law is substituted.

2.1.3 Magnetic input and the power balance A transverse field B_y gives the normalized input $u = B_y/B_0$ and interaction potential $-u\sin(d^{\mathsf{T}}q)$. Treating this interaction as external work yields the single physical port

$$\begin{aligned} g(q) &= d\cos(d^{\mathsf{T}}q), \\ y &= g(q)^{\mathsf{T}} v, \\ q' &= \partial_p H_*, \\ p' &= -\partial_q H_* - R_* v + g(q) u. \end{aligned} \tag{6}$$

Consequently $H_*' = -v^{\mathsf{T}} R_* v + uy$ and the SI power is $E_0\omega_0 H_*' = -\dot{q}^{\mathsf{T}} C_{\mathrm{phys}} \dot{q} + \mu B_y \cos(d^{\mathsf{T}}q) d^{\mathsf{T}}\dot{q}$. The distributed-viscosity identification trajectories use $u = 0$; the forced validation and feedback runs apply prescribed or held inputs through this same port. COMSOL Global Equations integrates these four reduced canonical equations in SI time. It does not provide a three-dimensional finite-element rod solution or a PDE-convergence certificate. When a conservative KAM statement is invoked, it concerns the subsystem obtained by setting both the input and dissipation to zero while retaining the same metric and stored potential. The dissipative training reference and the forced control system have their own power and stability statements; conservation is not inferred for either of those flows.

2.2 Complex energy and Rayleigh learning

The prescribed metric is shared by the simulator and learned model. With $s = d^{\mathsf{T}}q$, the learned potential is

$$\widehat{V}(q) = \tfrac{1}{2} q^{\mathsf{T}} B q + \tfrac{1}{2} s^2 - \tfrac{1}{24} s^4 + \eta_V \sum_{a=0}^{6} q_1^a q_2^{6-a} c_a^{\mathrm{ev}}(q), \tag{7}$$

where $\eta_V = \mathrm{binary64}(10^{-5})$ is interpreted as its exact stored value. Each c_a^{ev} is a real readout, symmetrized under $q \mapsto -q$, of two width-two complex-tanh hidden layers. The seven coefficient networks have 252 trainable real coordinates. They learn a sixth-and-higher-order correction and preserve the prescribed quadratic and quartic potential terms. Thus $\widehat{H} = p^{\mathsf{T}} G^{-1} p/2 + \widehat{V}$ is a trained mechanical Hamiltonian with known inertia; it is not an unrestricted identification of the complete energy.

A separate network has two width-two complex-tanh hidden layers and an affine complex readout forming a lower triangular factor $L_\theta(q)$. Its 42 raw real coordinates include real and imaginary parts. Define

$$\begin{aligned} \widehat{M} &= \mathrm{Re}(L_\theta L_\theta^*), \\ \widehat{R} &= S\widehat{M}S^{\mathsf{T}}, \\ \widehat{\mathcal{D}}(q,v) &= \tfrac{1}{2} v^{\mathsf{T}} \widehat{R}(q) v. \end{aligned} \tag{8}$$

For real z, $z^{\mathsf{T}}\widehat{M}z = \|L_\theta^* z\|_2^2 \geq 0$. The configuration-only dependence is essential: $\partial_v \widehat{\mathcal{D}} = \widehat{R}v$. The learned canonical equations therefore use $\widehat{H}$, $-\partial_v\widehat{\mathcal{D}}$ and the explicit magnetic port. The notation $\widehat{H} + i\widehat{\mathcal{D}}$ records these two real functions; the equations follow this decoder rather than a holomorphic-generator assumption. This separates the scalar-energy principle (Greydanus et al. 2019) from dissipative and input structure (Zhong et al. 2020; Desai et al. 2021).

2.2.1 Complex constraints and the actual weak objective Complex affine maps are evaluated by their real and imaginary components (Trabelsi et al. 2018). For the matrix head's literal complex tanh, the two constrained imaginary row budgets are each $1/2 < \pi/4$. The conservative coefficient networks retain their separate stored budgets 0.15 and 0.6, also below $\pi/4$. The first-layer bound $|\tanh| \leq 1$ supplies the component bounds used in the second layer. Positive smooth constraint denominators exclude the activation poles on the admitted real input domain. Their stored constants, rather than rounded effective weights reconstructed by a different evaluator, define the function certified below.

Only native times, configuration, observed velocity and prescribed input enter the optimizer. For a window with positive quadrature Q_w at its recorded times, put $\Pi = Gv$ and

$$\begin{aligned} a_i(V) &= \tfrac{1}{2} v^{\mathsf{T}} G_i v - \partial_i V + u g_i, \\ [\mathcal{D}M]_{wi} &= Q_w [S M S^{\mathsf{T}} v]_i, \\ r_Q &= \mathcal{D}\widehat{M} - Q a(\widehat{V}) + \Delta\Pi. \end{aligned} \tag{9}$$

The inertial term uses $G_i = \partial_i G$. Acceleration, native momentum derivatives and reference damping values are not training labels. The conservative stage profiles a nonnegative scalar drag coefficient and optimizes the potential coordinates. The later matrix stage freezes that potential and fits the distinct distributed-viscosity model. A forty-coordinate linear potential nuisance is profiled in that stage, saved, and retained in subsequent error calculations. It is not added to the deployed potential.

The numerical residual r_f includes row scaling, quadrature and a fixed computed map $I - Q_f Q_f^{\mathsf{T}}$. It need not be an exact orthogonal projector. The varying objective is

$$F(\theta) = \|r_f(\theta)\|_2^2 + \tfrac{\lambda}{2}\|\theta\|_2^2, \qquad \nabla F = 2J^{\mathsf{T}} r_f + \lambda\theta. \tag{10}$$

A fixed kinematic contribution is reported separately. Full-batch Gauss–Newton directions and Armijo acceptance operate on the real coordinates, with the Jacobian obtained by forward automatic differentiation. Initialization uses training weak equations, followed by PSD projection and a Cholesky factor; it does not use the reference matrix or its polynomial coefficients.

2.2.2 The optimization conclusion and its limits For $z = x + iy$ and a real objective, $\partial_{\bar z} F = (F_x + iF_y)/2$ and $\mathrm{d}F[p_z] = 2\,\mathrm{Re}[(\partial_{\bar z} F)^* p_z]$. The constrained objective has a real-analytic lift on its admitted finite parameter domain. In an exact-arithmetic augmented least-squares idealization with positive diagonal scale D_o, the direction satisfies

$$p = -B_o^{-1}\nabla F, \quad B_o = 2J^{\mathsf{T}}J + (\lambda+\eta)I + \mu D_o^{-2} \succ 0, \tag{11}$$

where $\lambda > 0$ and $\eta, \mu \geq 0$. It is a strict descent direction unless $\nabla F = 0$. The penalty makes the initial objective sublevel compact. Uniform spectral and gradient-Lipschitz bounds on a larger compact set give a positive lower bound for accepted step sizes under unlimited exact Armijo backtracking. Summing its decrease implies $\|\nabla F(\theta_k)\| \to 0$ for an infinite such iteration. Supplementary Section S2.2 proves this statement and distinguishes it from rounded least-squares solves and finite backtracking.

The strict complex-descent component motivated by Zhang et al. (2022, Theorem 3) is proved separately in the supplementary material. That fixed-gradient theorem does not establish convergence of the current Gauss–Newton implementation. Neither result proves global minimization, unique network weights or a physical prediction bound. The deterministic certificates below apply to the stored output whether or not a stationary point has been reached.

2.3 Complex canonical form and a symplectic difference scheme

The complex representation and its time discretization must retain the same canonical mechanics as the learned model. We first derive the complex equation from the real scalar Hamiltonian, then construct an implicit midpoint step for its nonseparable conservative field. The midpoint rule is classical (Hairer 2010, Lecture 2, Theorem 3); the derivation below specifies its application to the prescribed inertia and the deployed potential.

2.3.1 Wirtinger derivatives of the mechanical Hamiltonian All derivatives here use the normalized time τ. Set

$$\begin{aligned} z &= (q + ip)/\sqrt{2}, \qquad \bar z = (q - ip)/\sqrt{2}, \\ q &= (z + \bar z)/\sqrt{2}, \qquad p = (z - \bar z)/(i\sqrt{2}). \end{aligned} \tag{12}$$

On the physical real phase space, $\bar z$ is the conjugate of z. The Wirtinger derivatives treat these two arguments independently when applying the chain rule. For $\mathcal{H}(z,\bar z) = \widehat{H}(q(z,\bar z), p(z,\bar z))$, this gives

$$\partial_z \mathcal{H} = \frac{\nabla_q \widehat{H} - i\nabla_p \widehat{H}}{\sqrt{2}}, \qquad \partial_{\bar z}\mathcal{H} = \frac{\nabla_q \widehat{H} + i\nabla_p \widehat{H}}{\sqrt{2}}. \tag{13}$$

Consequently the canonical equations yield

$$z' = \frac{q' + ip'}{\sqrt{2}} = \frac{\nabla_p \widehat{H} - i\nabla_q \widehat{H}}{\sqrt{2}} = -i\,\partial_{\bar z}\mathcal{H}. \tag{14}$$

The normalization $1/\sqrt{2}$ fixes this coefficient. The corresponding real symplectic form is $\sum_j dq_j \wedge dp_j = i\sum_j dz_j \wedge d\bar z_j$.

For the actual nonseparable model, let $U(q) = G(q)^{-1}$ and $G_j = \partial_j G$. The identities

$$\begin{aligned} \partial_j U &= -U G_j U, \qquad v = U(q)p, \\ \widehat{H} &= \tfrac{1}{2} p^{\mathsf{T}} U(q) p + \widehat{V}(q). \end{aligned} \tag{15}$$

give the component derivatives

$$\partial_{p_j}\widehat{H} = v_j, \qquad \partial_{q_j}\widehat{H} = -\tfrac{1}{2} v^{\mathsf{T}} G_j v + \partial_j \widehat{V}. \tag{16}$$

Thus $\mathcal{H}$ is explicitly

$$\mathcal{H}(z,\bar z)=\tfrac12 p(z,\bar z)^{\mathsf T}U(q(z,\bar z))p(z,\bar z)+\widehat V(q(z,\bar z)),$$

and equations (13)–(16) retain the configuration derivative of inertia. They do not assume $\widehat H=T(p)+\widehat V(q)$.

The real scalar energy is generally nonholomorphic in the physical variable z: its dependence on $\bar z$ is precisely what appears in (14). This fact does not conflict with the real-analytic complexification required by KAM. In that calculation, real canonical variables extend to independent complex variables (Q,P), and the expressions $(Q+iP)/\sqrt2$ and $(Q-iP)/\sqrt2$ are independent linear coordinates, not a conjugate pair. For the actual energy in (7), define

$$\begin{aligned}V_4(Q)&=\tfrac12 Q^{\mathsf T}BQ+\tfrac12(d^{\mathsf T}Q)^2-\tfrac{1}{24}(d^{\mathsf T}Q)^4,\\ c_a^{\mathrm{ev},\mathbb C}(Q)&=\frac{F_a(Q)+F_a(-Q)+F_a^{\#}(Q)+F_a^{\#}(-Q)}{4},\\ \widehat V^{\mathbb C}(Q)&=V_4(Q)+\eta_V\sum_{a=0}^{6}Q_1^aQ_2^{6-a}c_a^{\mathrm{ev},\mathbb C}(Q),\end{aligned}\tag{17}$$

where $F_a^{\#}(Q)=\overline{F_a(\bar Q)}$ conjugates the stored network coefficients and η_V is the same exact stored amplitude. On real arguments this recovers the deployed real readout and its even symmetrization. Taking $\operatorname{Re}F_a(Q)$ at complex arguments would not provide that holomorphic extension. The required domain also excludes the network's $\tanh$ poles and singularities of G^{-1}. Analytic continuation of the metric retains its ordinary matrix transpose; it does not introduce a Hermitian Gram product.

For the configuration-only Rayleigh matrix $R(q)$, set $\mathcal D(q,v)=v^{\mathsf T}R(q)v/2$. At fixed q, symmetry gives $\partial_v\mathcal D=R(q)v$. Adding the magnetic port $g(q)=d\cos(d^{\mathsf T}q)$ to the momentum equation therefore gives

$$z'=-i\,\partial_{\bar z}\mathcal H+\frac{i}{\sqrt2}\big[-R(q)v+ug(q)\big].\tag{18}$$

This is the complex form of the same dissipative forced mechanics; the additional term is not absorbed into a holomorphic scalar energy. The symplectic construction below concerns the conservative subsystem with $R=0$ and $u=0$.

2.3.2 Implicit midpoint for the same nonseparable energy Write $x=(q,p)$ and

$$J_c=\begin{pmatrix}0&I_2\\-I_2&0\end{pmatrix},\qquad f(x)=J_c\nabla\widehat H(x).$$

Applying one midpoint evaluation to $x'=f(x)$ defines

$$x_{n+1}=x_n+hJ_c\nabla\widehat H(m_n),\qquad m_n=(x_n+x_{n+1})/2.\tag{19}$$

Using (16), its actual coupled equations are

$$\begin{aligned}\bar q&=(q_n+q_{n+1})/2,\\ \bar p&=(p_n+p_{n+1})/2,\qquad \bar v=U(\bar q)\bar p,\\ q_{n+1}&=q_n+h\bar v,\\ p_{n+1,j}&=p_{n,j}+h\big[\tfrac12\bar v^{\mathsf T}G_j(\bar q)\bar v-\partial_j\widehat V(\bar q)\big].\end{aligned}\tag{20}$$

Equivalently, the same affine coordinate change gives

$$z_{n+1}=z_n-ih\,\partial_{\bar z}\mathcal H\left(\frac{z_n+z_{n+1}}{2},\frac{\bar z_n+\bar z_{n+1}}{2}\right).$$

This is an implicit conservative integrator. It does not reclassify the documented native collection or held-flow solvers as midpoint schemes.

To specify local solvability, let $\overline B(x,b)$ lie inside a real domain where $G\succ0$, the potential is C^3, and $\|f\|\le M$, $\|Df\|\le L$. If

$$|h|M<b,\qquad |h|L/2<1,\tag{21}$$

then $y\mapsto x+hf((x+y)/2)$ maps the closed ball into itself and is a contraction. Its unique fixed point defines $\Phi_h(x)$ there. The Jacobian $I-hDf/2$ is invertible, so the implicit-function theorem gives a differentiable local step map. Uniform compact-domain versions of these conditions are used when composing steps.

Proposition 1. Symplectic midpoint map. *Under* (21)*, the exact local solution map of* (19) *satisfies* $(D\Phi_h)^{\mathsf T}J_cD\Phi_h=J_c$.

Proof. At the midpoint let $\mathsf H_m=\nabla^2\widehat H(m_n)$ and $A=J_c\mathsf H_m$. Since $\mathsf H_m^{\mathsf T}=\mathsf H_m$ and $J_c^2=-I$,

$$A^{\mathsf T}J_c+J_cA=0.\tag{22}$$

Differentiating the implicit step with respect to x_n gives

$$(I-hA/2)D\Phi_h=I+hA/2.$$

Set $U_h=I-hA/2$ and $V_h=I+hA/2$. Equation (22) implies

$$V_h^{\mathsf T}J_cV_h=J_c+(h^2/4)A^{\mathsf T}J_cA=U_h^{\mathsf T}J_cU_h.$$

The matrices U_h,V_h commute because both are polynomials in A; their inverses commute whenever defined. Thus

$D\Phi_h = U_h^{-1}V_h = V_hU_h^{-1}$ and

$$(D\Phi_h)^\mathsf{T} J_c D\Phi_h = U_h^{-\mathsf{T}} V_h^\mathsf{T} J_c V_h U_h^{-1} = J_c.$$

The argument allows all configuration-dependent inertia and potential terms inside the symmetric Hessian H_m.

2.3.3 Second-order accuracy and the scope of preservation Suppose a common convex neighborhood contains all exact and numerical trajectory states and all segments used in the comparisons below, with $\|f\| \le M$, $\|Df\| \le L$ and $\left\|D^2 f\right\| \le B_2$. These bounds follow locally from C^3 Hamiltonian regularity on a compact pole-free region. Fix $h_0 > 0$ with $\kappa = h_0 L/2 < 1$, and define

$$C_d = \frac{B_2 M^2}{24} + \frac{L^2 M}{6}, \qquad C_\ell = \frac{C_d}{1-\kappa}, \qquad \Lambda = \frac{L}{1-\kappa}.$$

For $0 < h \le h_0$, the one-step error against the exact conservative flow is at most $C_\ell h^3$. With the same exact initial state and $nh \le T$, the accumulated error satisfies

$$\|x_n - x(nh)\| \le C_\ell h^2 \frac{e^{\Lambda T} - 1}{\Lambda}, \tag{23}$$

where the quotient is T when $\Lambda = 0$. The derivation, including the required common-domain condition, appears in Supplementary Section I.

Symplecticity concerns the exactly solved conservative step. A finite nonlinear-solver residual r_n contributes a state error at most $\|r_n\| / (1 - |h|L/2)$; unverified tolerances do not establish exact symplecticity. For a quadratic Hamiltonian the exact identity

$$\widehat{H}(x_{n+1}) - \widehat{H}(x_n) = \nabla \widehat{H}(m_n)^\mathsf{T}(x_{n+1} - x_n) = 0$$

follows from $a^\mathsf{T} J_c a = 0$. A nonquadratic $\widehat{H}$ generally has an additional energy increment, so the identity does not imply exact energy preservation for the deployed network. The added Rayleigh field has divergence $-\operatorname{tr}(RG^{-1})$. For zero or externally prescribed input, a positive trace therefore implies canonical-volume contraction and excludes symplecticity of the dissipative flow. A state-dependent feedback $u(q,p)$ contributes the additional divergence $g(q)^\mathsf{T}\nabla_p u$. Finally, second-order consistency and a symplectic step do not prove a KAM torus for the discrete map: step-size nonresonance, map geometry and an appropriate invariance-defect theorem would be additional requirements. The article's certified KAM endpoint remains the continuous conservative Hamiltonian flow.

3 Numerical approximation and learning-error analysis

3.1 Midpoint integration of the learned conservative field

The coupled implicit midpoint equations were evaluated for the same learned Hamiltonian used in the conservative verification. This calculation retains its saved finite-Gauss inertia, quartic prior and trained complex potential correction, with dissipation and input set to zero. The initial normalized canonical state was $(q_1, q_2, p_1, p_2) = (0.03, 0.02, 0.002, -0.0015)$, and the final time was $\tau = 1$. The initial state, step sizes, Newton stopping rule and comparison settings were fixed before the numerical run.

Each step solves the four coupled equations by Newton iteration with the automatic-differentiation Jacobian of the same conservative field. The residual stopping threshold was 2×10^{-15} in the maximum norm, with a maximum of twelve updates. All 140 steps converged using two updates. Table 1 compares the results with DOP853 applied directly to that learned field. The reference used relative tolerance 2.5×10^{-14}, absolute tolerance 5×10^{-17} and maximum step $1/160$. A second reference with absolute tolerance 10^{-16} and maximum step $1/80$ differed by at most 5.79×10^{-17} at the common comparison nodes.

The two error ratios approach four, as expected from second-order convergence. The nonzero energy increments are consistent with applying midpoint to a nonquadratic Hamiltonian. At every converged midpoint, the field Jacobian obtained by automatic differentiation was used in the Cayley expression for the step Jacobian. The largest observed entry of $D^\mathsf{T} J_c D - J_c$ was below 4.45×10^{-16}. For each step size, an additional derivative probe differentiated through the actual two Newton updates at the initial state. Its state output matched the executed first step bit-for-bit; its largest symplectic defect was below 2.23×10^{-16}, and its Jacobian differed from the Cayley value by at most 1.67×10^{-16}.

The monitored step boundaries and Newton midpoint iterates remained within $|x_j| < 0.030067$. At those points the minimum metric eigenvalue exceeded 0.162619, and the largest evaluated quantity $h\|Df\|_2/2$ was below 0.153804. These observations concern the evaluated points and provide no uniform bound on a surrounding ball. The convergence and symplectic-defect measurements verify the numerical interface of the stated algorithm; the exact-step symplectic theorem retains its domain and exact-solve hypotheses. The calculation supplies neither an exact symplecticity certificate

Table 1. Float64 midpoint results for the unchanged learned nonseparable Hamiltonian. State errors use the Euclidean norm in normalized canonical coordinates; the maximum is over the computed step boundaries. The ratio compares the error at h with that at $h/2$. Energy columns report measured excursions and increments of this same learned energy. These are numerical diagnostics against a numerical reference, rather than rigorous flow enclosures.

h	Final error	Maximum node error	Ratio	$\max_n \lvert H_n - H_0\rvert$	$\max_n \lvert H_{n+1} - H_n\rvert$	Newton residual
$1/20$	9.58774×10^{-4}	9.58774×10^{-4}	3.95783	1.01842×10^{-9}	3.29647×10^{-10}	6.08×10^{-18}
$1/40$	2.42248×10^{-4}	2.42248×10^{-4}	3.98939	2.55850×10^{-10}	4.45323×10^{-11}	2.17×10^{-18}
$1/80$	6.07231×10^{-5}	6.07231×10^{-5}	—	6.40440×10^{-11}	5.59844×10^{-12}	2.06×10^{-18}

for floating-point Newton iterations nor a discrete-map KAM theorem.

3.2 From native observations to deployed-function error

Finite weak residuals require a quantitative inverse before they can control a function on a domain. Here the reference class also specifies the known metric, the low-order potential jet and independent analytic remainder bounds. These class assumptions are not learned from a rank calculation.

Let χ contain the forty normalized monomials of total degrees 6, 8, 10 and 12, with normalization $r_0 = \mathrm{binary64}(0.1)$ interpreted exactly. The matrix representative $P_4(c)$ contains all fifteen monomials of degree at most four in each of three symmetric coordinates, giving 45 coordinates. Decompose the frozen functions as

$$\begin{aligned} \widehat{V} - V_* &= \chi^{\mathsf{T}} a_V + \rho_V, \\ \widehat{M} &= P_4(\widehat{c}) + \widehat{\rho}_M, \qquad M_* = P_4(c_*) + \rho_{M,*}. \end{aligned} \tag{24}$$

These analysis representatives do not change either deployed network. Fourteen trajectories provide 560 momentum-window components. Write $C = [N, A] \in \mathbb{R}^{560\times 85}$, where $N = Q\nabla\chi$ and A is the matrix-polynomial weak design. The full saved nuisance is b, so $r_b = r_Q + Nb$. Set $\widehat{t} = \mathcal{D}\widehat{\rho}_M$, $t_* = \mathcal{D}\rho_{M,*}$, $t_V = Q\nabla\rho_V$ and $e = Qa(V_*) - \mathcal{D}M_* - \Delta\Pi$. Subtraction of the native equations gives the signed identity

$$r_b = A(\widehat{c} - c_*) + N(a_V + b) + \widehat{t} - t_* + t_V - e. \tag{25}$$

The consistency term e accounts for the actual quadrature and recorded observations. Independently enclosed reference IVPs establish the true-flow observation errors; a native solver tolerance is not used as that certificate.

For fixed positive column scaling D_c, let L_c be a numerical left inverse of CD_c^{-1}. An outward bound for $T_c = L_c C D_c^{-1}$ proves $\|T_c - I\|_\infty < 1$. The corrected inverse $K = D_c^{-1} T_c^{-1} L_c$ consequently obeys $KC = I_{85}$. Its first forty rows K_V yield

$$a_V = K_V(r_b - \widehat{t}) - b + K_V e + K_V t_* - K_V t_V. \tag{26}$$

Subtracting b is necessary: even with zero tails and consistency error, $b = -a_V$ can make the profiled residual vanish while the deployed potential still differs from the reference.

Theorem 2. Fixed-design conservative error. *For a real or complex domain $\mathcal{O}$ and a multi-index α, suppose*

$$\begin{aligned} B_\alpha &\ge \sup_{\mathcal{O}} \lvert(\partial^\alpha\chi)^{\mathsf{T}}[K_V(r_b - \widehat{t}) - b]\rvert, \\ \beta_\alpha &\ge \sup_{\mathcal{O}} \left\|(\partial^\alpha\chi)^{\mathsf{T}} K_V\right\|_2 . \end{aligned}$$

If $\|e\|_2 \le E$, $\|t_*\|_2 \le T_*$, $\|t_V\|_2 \le V_w$ and $\|\partial^\alpha\rho_V\|_{\infty,\mathcal{O}} \le R_\alpha$, then

$$\left\|\partial^\alpha(\widehat{H} - H_*)\right\|_{\infty,\mathcal{O}} \le B_\alpha + \beta_\alpha(E + T_* + V_w) + R_\alpha. \tag{27}$$

Indeed, multiplying (26) by $(\partial^\alpha\chi)^{\mathsf{T}}$ and applying Cauchy–Schwarz proves the statement because the kinetic terms agree. Signed Bernstein polynomial bounds and independent analytic tails are used rather than absolute coefficient sums alone. Finite Taylor identities transport the real derivative bounds to a complex rectangle. The last 45 rows of K, with their own output gains, similarly control $\widehat{M} - M_*$. The full derivation is in Supplementary equations (S10)–(S13).

3.2.1 A sufficient neighborhood of the observed design

All fourteen nominal training IVPs are independently enclosed. Uniform feature derivatives transport their state uncertainty to a scaled design perturbation. For one future trajectory in each of the same fourteen directions, the certified sufficient conditions are an initial-state perturbation at most 2^{-34} in the specified mechanical W norm and paired configuration/velocity observation errors at most 10^{-11}, at unchanged stored times and quadrature. The total scaled perturbation is below $0.039651094 < 1/8$, so the corrected inverse persists by a Neumann argument (Supplementary Appendix A). This is a positive local design radius, not a minimum sample count or a guarantee for arbitrary fourteen trajectories. Residuals, the finite nuisance and analytic remainders remain required for a new trained output.

It concerns observability of the finite representatives, not identifiability of all network parameters.

3.2.2 Point-gradient validation without weak-loss profiling A separate validation design retains both gradient components at forty configurations. It uses the same forty-dimensional potential representative, with an 80×40 matrix A_{pt} formed from $\nabla\chi(Q_i)$. For labels Y, let

$$z = \nabla\widehat{V}(Q) - Y,$$
$$\xi = Y - \nabla V_*(Q), \qquad t = \nabla\rho_V(Q).$$

An exact corrected left inverse $J_{\mathrm{pt}}A_{\mathrm{pt}} = I$ gives

$$a_V = J_{\mathrm{pt}}z + J_{\mathrm{pt}}(\xi - t). \tag{28}$$

This pointwise identity does not profile a new potential nuisance or alter the deployed Hamiltonian. Supplementary Proposition S10 gives its signed-principal, label-error and analytic-tail bounds.

The inverse and label-error contract are common to the candidate outputs. Supplementary Theorem S5 proves simultaneous function-error bounds for outputs trained or selected using these same observations, with each candidate supplying its own valid observation-point and complex-domain remainder certificates. The measured residual may be nonzero. Its model-dependent embedding and KAM witness remain separate inputs to the conservative verification.

The configurations follow a once-drawn finite-bit protocol, with all forty records retained in their original order. Exact rational arithmetic certifies the realized matrix inverse. Under the independently uniform bit model, a polynomial witness separately bounds rank failure by $340/2^{48}$; it supplies no conditioning or accuracy guarantee. The associated finite-Gauss simulation protocol uses $p(0) = u = 0$ and the field-defined velocity observer $v = G(q)^{-1}p$. Actual recorded SI times determine exact Lagrange derivative weights for labels reconstructed from $G(q)v$. Native momentum, stored derivatives and forces remain diagnostic records. The protocol seals every observation-only label before reference-gradient comparison. The realized label-error, signed-principal and complete same-pair acceptance conditions are reported in Section 3.3.5.

The Hamiltonian remains frozen throughout this validation. The sampling units are forty configurations; their five-node trajectories give 200 observations, which are not 200 iid state samples. The numerical protocol uses $h = 2^{-27}$ and total normalized horizon $4h = 2^{-25}$. With $\omega_0 \simeq 11.4172873\,\mathrm{s}^{-1}$, the SI horizon is approximately 2.61×10^{-9} s. This short window is chosen for the strict truncation budget. Internal latent momentum generates the simulated velocity but is excluded from label files and reconstruction. Physical deployment would require separate velocity-error and clock bounds and a calibrated drag law; the viscosity used here remains the declared numerical assumption.

3.3 Numerical results

3.3.1 Distinct conservative and dissipative training cohorts The numerical reference is the reduced finite-Gauss mechanical system of Section 2.1. Two distinct COMSOL Global Equations cohorts each contain fourteen training trajectories and four reserved trajectories, with 801 stored nodes per trajectory. Table 2 distinguishes their observation horizons and learning tasks. These are prescribed trajectory designs. The training inputs are normalized time, configuration, observed velocity and applied input; native momentum, reference forces and reference damping matrices are excluded from the optimization labels.

For conservative learning, the inertia, bending term and potential through degree four are prescribed. Seven even coefficient functions, represented by two-hidden-layer complex networks, multiply the degree-six monomials. The full-cosine higher-order potential is deliberately hidden from the optimizer, defining a controlled identification benchmark. A nonnegative scalar mass-proportional drag is profiled from the training weak equations, including during initialization. The potential receives 400 accepted full-batch Armijo steps. Its final weighted, scaled momentum-weak loss is 1.669482×10^{-10}; the fixed kinematic term and parameter penalty are separate contributions. This finite nonzero value does not establish exact recovery or convergence to a global optimum.

The distributed-viscosity cohort is generated separately. The previously trained Hamiltonian is frozen while the complex matrix head is fitted. Twenty windows per training trajectory provide 560 momentum components. Positive five-point Newton–Cotes weights use the actual stored time labels. Forty linear potential-nuisance coordinates are profiled and saved for error assessment, without modifying the deployed conservative potential. The procedure therefore learns the conservative correction and matrix dissipation sequentially; the earlier scalar drag and the later distributed matrix law have distinct roles.

3.3.2 Comparator behavior and uniform matrix-error bounds Table 3 compares a constant factor, a real-valued factor network and the complex factor network.

Table 2. Separate training cohorts and the fitted complex models. Coordinate counts refer to stored trainable real parameter coordinates in each stage.

Stage	Train/reserved	Time interval	Stored real coordinates	Accepted steps
Potential $\widehat{V}$	$14/4$	$[0, 0.2]$	252	400
Matrix $\widehat{M}$	$14/4$	$[0, 0.4]$	42	100

A common Gauss–Newton/Armijo budget permits 100 accepted steps. The constant and real models terminate after two and fifteen accepted steps, respectively, when their line searches fail; the last accepted parameters are retained. The complex model completes the budget. Training-only protocol refinement precedes the final model freeze. The four reserved trajectories are then evaluated once, transferring the training nuisance coefficients unchanged and fitting no additional holdout coefficients.

Both state-dependent models improve these metrics over the constant baseline, and the real model performs better than the complex model. The real and complex heads share the data, PSD decoder, loss and budget, but the real model includes a quadratic factor skip and has a different topology. Equal parameter counts consequently support a limited comparison, rather than an isolated test of complex arithmetic or a claim of complex-network superiority.

Positive semidefiniteness is imposed by the factor decoder in every fit. The comparison therefore measures identification within an already dissipative model class. It does not test whether unconstrained networks happen to learn the power sign. The conservative checkpoint is identical across the three fits, so their different weak losses do not represent different learned Hamiltonians.

The grid is a post-fit diagnostic. A separate outward-rounded verification proves an operator-error upper bound of 2.822031×10^{-9} for the frozen complex head against the full reference on $|q_i| \le 0.1$. The observation-to-output construction of Section 3.2 instead combines an observation-computable principal term with independent consistency and remainder bounds, obtaining the tighter upper bound 2.718680×10^{-9} on that box. This second conclusion retains the learned-head tail, reference tail and potential-nuisance remainder; it is distinct from direct post-fit function comparison. The direct grid comparison is computed after training freezes. Exact certificate bounds are retained separately from their displayed decimals; loss values and grid maxima remain rounded numerical diagnostics.

3.3.3 Conservative certification and a nearby-design budget The finite-design conservative certificate concerns the same frozen potential, including subtraction of the saved nuisance coordinate. On $|q_i| \le 0.06$, its energy-error upper bound is 5.909451×10^{-14}. On the complex configuration rectangle with real half-width 0.06 and imaginary half-width 0.01, the componentwise first-derivative bound is below 8.324054×10^{-12}. These model-error bounds enter the KAM transport, whereas the corrected embedding requires its own invariance-residual estimate. The complete strip residual is below 1.205897×10^{-41}, beneath the sufficient threshold 5.82604×10^{-38}. The conclusion concerns the stated conservative learned–reference family and prescribed frequency.

The fourteen reference training IVPs are also enclosed at all 11,214 stored nodes and across 11,200 integration slabs. Combining their true-flow observation errors with uniform feature derivatives gives a local fourteen-direction design condition. An initial perturbation of at most 2^{-34} in the specified W norm, paired observation errors at most 10^{-11}, and unchanged stored clocks and quadrature imply a scaled design perturbation below $0.039651094 < 1/8$. This is a sufficient condition for future accepted bundles. No perturbed bundle has been generated or fitted; residual, nuisance, analytic-tail and KAM acceptance conditions remain necessary for a new learned output.

3.3.4 Population-risk integration and point-design conditioning For the known frozen conservative pair, outward integration of the sixth power of each polynomial gradient component, followed by the independent analytic-tail allowance, gives $R_6 < 2.395279 \times 10^{-14}$. The corresponding sufficient factor-512 risk-root allowance exceeds 1.012617×10^{-13}. The transported complex partial bounds of orders one, two and three are below 1.483543×10^{-10}, 4.015155×10^{-8} and 8.988099×10^{-6}, respectively. The separately recomputed eight same-pair KAM gates all pass; their largest value is below 2.385535×10^{-4}. These bounds arise from known-reference function integration. They are not estimates of population risk from native samples.

3.3.5 Fresh point observations and conservative error acceptance The separately frozen forty-point design is evaluated through five-node rest-start COMSOL trajectories, with the conservative checkpoint fixed. All forty configurations are retained in their original random-record order. The configurations are the sampling units; the 200 stored trajectory nodes are correlated within each short observation

Table 3. Dissipation-head comparison. L_{train} is the scaled projected momentum loss; R_{hold} is the unscaled weak-residual ℓ^2 norm with the transferred training nuisance. The grid column is the maximum operator error against the reference quartic approximation on an 81×81 configuration grid. These three columns have different meanings.

Head	Real coordinates	Accepted steps	L_{train}	R_{hold}	Grid maximum
Constant	3	2	6.768681×10^{-2}	3.873143×10^{-11}	4.240111×10^{-8}
Real	42	15	5.480229×10^{-9}	1.582292×10^{-14}	7.343073×10^{-11}
Complex	42	100	3.805744×10^{-6}	1.737922×10^{-12}	2.247102×10^{-9}

window. The field-defined velocity observer uses the internal simulated momentum through $G(q)^{-1}p$, while the label reconstruction reads only observed configuration, velocity, input and exact recorded clocks. Native momentum, stored derivatives and forces are confined to diagnostic exports. The normalized horizon is $4h = 2^{-25}$ with $h = 2^{-27}$, corresponding to approximately 2.61×10^{-9} s at the prescribed time scale.

All forty observation-only labels are sealed before the known-reference comparison. That comparison gives

$$\begin{aligned} \max_{i,j} |\xi_{ij}| &\le 7.850603 \times 10^{-16}, \\ \|\xi\|_2 &\le 2.040666 \times 10^{-15}. \end{aligned} \tag{29}$$

where $\xi_{ij} = Y_{ij} - \partial_j V_*(Q_i)$. These controlled-reference fidelity bounds satisfy the predeclared component and vector capacities. They concern the realized simulation observations, without assigning an independent physical-sensor or unknown-plant error law.

The 80×40 design has an exact corrected left inverse, with $\|I - RA_{\mathrm{pt}}\|_\infty < 1.410 \times 10^{-47}$. The signed observable principal uses only $z = \nabla\widehat{V}(Q) - Y$. Correcting Rz to $J_{\mathrm{pt}}z$, adding the controlled label-error allowance, and retaining both point and final analytic tails verifies all nine partial-error budgets through order three. Their orderwise maxima are bounded by

$$\begin{aligned} E_1^{\mathrm{obs}} &\le 2.025150 \times 10^{-10}, \\ E_2^{\mathrm{obs}} &\le 1.860781 \times 10^{-8}, \\ E_3^{\mathrm{obs}} &\le 1.740994 \times 10^{-6}. \end{aligned} \tag{30}$$

The four ordered field increments and all eight recomputed same-pair KAM inequalities satisfy their declared bounds; the largest scalar gate is below 2.205303×10^{-4}. The complete embedding residual remains below 1.205897×10^{-41}. Supplementary Section F.3 gives the component budgets. The calculation retains the same endpoints, frequency, cubic response, Fourier data and large-domain analytic premises as the earlier proof. The oracle population-risk calculation above is a separate verification; these point-observation bounds are not substituted for that risk.

3.3.6 Ablation of the learned conservative correction A separate comparison evaluates two frozen outputs against the same sealed labels: the prescribed quadratic–quartic prior V_4 and the learned $\widehat{V}$. Both retain the identical inertia, bending and magnetic coefficients. Setting the learned higher-order correction to zero defines the prior; no model is refitted on these forty configurations, and no reference gradient or label-noise diagnostic enters this comparison. Define the gradient-vector RMSE and empirical sixth-root statistic by

$$\mathrm{RMSE}(V) = \left[\frac{1}{40} \sum_{i=1}^{40} \sum_{j=1}^{2} |\partial_j V(Q_i) - Y_{ij}|^2 \right]^{1/2},$$

$$\widehat{r}_6(V) = \left[\frac{1}{40} \sum_{i=1}^{40} \sum_{j=1}^{2} |\partial_j V(Q_i) - Y_{ij}|^6 \right]^{1/6}.$$

The normalization is per configuration, with both gradient components included. The latter statistic is a sixth root, not a sixth-power population risk.

The RMSE enclosures strictly separate, with a prior-to-learned ratio of approximately 5298.34. This supports the usefulness of the learned higher-order correction for this frozen benchmark and label protocol. It is a correction ablation, distinct from the real/complex and prior/no-prior architectural comparisons.

3.3.7 Generic phase-space networks: representation and quadratic prior A separate factorial experiment compares real and complex networks with and without a fixed quadratic Hamiltonian. It uses the current finite-Gauss mechanics and distributed-viscosity observations. The four cells are R (real, no fixed quadratic term), C (complex, no fixed quadratic term), GR (real with the quadratic prior), and GC (complex with the quadratic prior). The quadratic prior improves both paired representations, and GC has the smallest factorial error for every seed. The absolute errors nevertheless remain substantially larger than those of the shared fourth-order model with the network correction set to zero.

Table 4. Frozen conservative-correction ablation on the same forty sealed point-gradient labels. Entries are rounded interval diagnostics. The comparison varies the learned correction within one mechanical model.

Output	Gradient-vector RMSE	$\widehat{r}_6$
Quadratic–quartic V_4	$\approx 1.297917 \times 10^{-11}$	$\approx 3.168798 \times 10^{-11}$
Learned $\widehat{V}$	$\approx 2.449669 \times 10^{-15}$	$\approx 5.265748 \times 10^{-15}$

Writing $G_0 = G(0)$ and $\mathsf{K}_2 = B + dd^T$, every cell has the form

$$\begin{aligned} H_s(q,p) &= T_{\mathrm{nl}}(q,p) + V_{\mathrm{quartic}}(q) \\ &\quad + I_s H_2(q,p) + N_{\theta,s}(q,p), \\ T_{\mathrm{nl}}(q,p) &= \tfrac{1}{2} p^T [G(q)^{-1} - G_0^{-1}] p, \\ V_{\mathrm{quartic}}(q) &= -\tfrac{1}{24} (d^T q)^4, \\ H_2(q,p) &= \tfrac{1}{2} p^T G_0^{-1} p + \tfrac{1}{2} q^T \mathsf{K}_2 q, \end{aligned} \tag{31}$$

where $I_s = 0$ for R and C and $I_s = 1$ for GR and GC. Thus the ablation removes the complete origin quadratic Hamiltonian, including its momentum block. The known nonlinear kinetic remainder and quartic magnetic term are common to all cells; R and C are not models without any prescribed mechanics.

The generic residual acts on all four canonical coordinates $x = (q,p)$. Its inputs use componentwise RMS scales computed from the fourteen training trajectories, and its energy scale is the training RMS of H_2. A common evenization and subtraction at the origin impose $N_{\theta,s}(-x) = N_{\theta,s}(x)$ and $N_{\theta,s}(0) = 0$. No degree-six prefactor or independent quadratic skip is present. In particular, the residual Hessian is free, so the no-prior cells are not structurally prevented from approximating the missing quadratic term. This is an approximation task for finite-width networks, not a claim of exact quadratic representation at arbitrary parameter values.

The real network has two width-24 hidden layers with real tanh and a linear readout, giving 744 trainable real coordinates. The complex network has two width-16 hidden layers with literal complex tanh, independent $z, \bar{z}$ affine inputs, and a real readout, giving 736 real coordinates. Its layerwise imaginary-argument budgets are 0.15 and 0.6. These constraints apply on the declared normalized phase box containing all training states. The counts differ by approximately 1.09%; there is no parameter padding. Widths, parameter domains and initialization laws also differ. Within a representation and seed, the prior and no-prior cells start with identical raw network parameters; real and complex initial functions are not matched.

All cells use the same frozen 42-coordinate complex PSD matrix head and the same distributed-viscosity cohort of fourteen trajectories on $[0, 0.4]$, with 801 observation nodes per trajectory. This is distinct from the conservative $[0, 0.2]$ cohort used to fit the certified 252-coordinate structured Hamiltonian. The matrix head is the previously identified distributed-viscosity head, fitted with that structured Hamiltonian frozen; it is not refitted for these comparators. Only time, configuration, observed velocity and prescribed input enter the fit. Observed momentum is reconstructed as $G(q)v$. The predicted dynamics use

$$\begin{aligned} q' &= H_{s,p}, \\ p' &= -H_{s,q} - \widehat{R}(q) H_{s,p} + u g(q). \end{aligned} \tag{32}$$

Consequently, dissipation uses the model's actual $H_{s,p}$, and the kinematic discrepancy is $H_{s,p} - v$; observed velocity is not silently substituted into the predicted damping force. Twenty windows per trajectory use the same actual-time, five-point Newton–Cotes quadrature. The common scalar loss is the mean normalized momentum-weak square plus 0.1 times the quadrature-weighted normalized kinematic square and a raw-coordinate penalty with coefficient $10^{-20}/2$. All scales come from the fourteen training trajectories. No separate potential-nuisance fit, reference-force labels or validation selection is used.

Seeds 43, 47 and 53 give twelve fits. Each completes 400 full-batch Adam updates with learning rate 0.01, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The lowest finite training total among the initial state and the 400 recorded updates selects each checkpoint. All twelve selected outputs are frozen before this factorial evaluation, and their training-loss replays agree. There are no failed fits, no failed forty-point model evaluations and no failed model–holdout evaluations among the 48 latter combinations. The three seeds are optimization repeats on common data, rather than independent physical datasets. The initial functions also differ materially between representations: for seed 43, the initial training totals for R, C, GR and GC are 1.197470, 1.085593, 0.139463 and 0.001348, respectively. The common learning rate and update budget therefore do not remove the effects of the different initialization and constraint-induced function scales, especially in the prior-assisted pair.

The primary evaluation uses the same forty sealed observation-only gradient labels Y_i at the prescribed rest

points $(Q_i, 0)$:

$$E_s^2 = \frac{1}{40}\sum_{i=1}^{40}\left\|H_{s,q}(Q_i,0) - Y_i\right\|_2^2. \tag{33}$$

The forty labels and four original holdout trajectories had already served other analyses. They are therefore reused validation, not a new blind confirmation of these generic comparators. At $p = 0$, the metric tests potential gradients; it does not validate arbitrary phase-space derivatives. Table 5 retains every seed.

The prior improves both representations in every paired seed, and the complex fit has lower RMSE within both prior settings. The interaction is negative for every seed: the prior produces the larger absolute error reduction for the real network and narrows the absolute real–complex gap. These observations describe this data, architecture pair and optimization budget. They do not isolate complex arithmetic from the different widths, constraints, initialization and parameter-dependent regularization, or establish universal complex superiority.

Full-phase consistency is assessed separately on the four reused holdout trajectories, using the fixed training scales, the same frozen matrix head and no additional coefficients. Table 7 pools squared normalized errors equally across the three seeds and four trajectories before taking the square root. It reports weak momentum and quadrature-weighted $H_{s,p} - v$ discrepancies, not a new integrated rollout. The last column gives the range across seeds of $\{40^{-1}\sum_i \|H_{s,p}(Q_i,0)\|_2^2\}^{1/2}$. Although the reference velocity vanishes at these rest points, the generic residual can contain mixed q, p terms. Central evenization does not impose evenness in p at fixed q, and the nonzero rest velocity diagnostic remains relevant even when the potential-gradient error is relatively small.

The absolute-accuracy limitation is substantial. On the identical forty labels, the zero-network fourth-order baseline $H_4 = T_{\mathrm{nl}} + H_2 + V_{\mathrm{quartic}}$ has gradient-vector RMSE approximately 1.297917×10^{-11}. Every generic GC fit is worse: its RMSE ranges from 7.966939×10^{-6} to 1.098578×10^{-5}, approximately 6.1×10^5 to 8.5×10^5 times the baseline error. The generic prior-assisted family contains H_4 at zero residual, so this gap is a limitation of the stated generic training protocol at the resolved signal scale, not proof that the physical prior is unrepresentable. Relative improvement within the factorial experiment therefore does not demonstrate improvement over the prescribed fourth-order physics. The structured higher-order correction ablation addresses that separate question and is retained independently.

These comparisons use ordinary floating-point network evaluation. Exact rational propagation of the sealed label widths supplies sensitivity bounds conditional on the returned network numbers; it does not enclose neural evaluation and differentiation errors and is not a strict network-ranking certificate. None of the twelve generic outputs replaces the certified 252-coordinate structured Hamiltonian or inherits its conservative KAM, learning-error or feedback certificates. The results establish a qualified factorial comparison and expose its absolute-accuracy limitation.

3.3.8 A precision-centered prior-assisted comparison A separate two-fit comparison tests whether objective and output scaling account for part of the absolute-accuracy gap in Section 3.3.7. It retains the generic full-phase real and complex architectures, with 744 and 736 trainable real coordinates, respectively, and uses seed 43 only. Both models now have the actual scalar form $H_4 + N_s$, with the same finite-Gauss kinetic energy and fourth-order potential. An exactly zero readout makes their initial functions identical to H_4; there is no degree-six multiplier, polynomial replacement or separately fitted potential block.

The output multipliers are $\rho_{\mathrm{GR}} = 2^{-32}$ and $\rho_{\mathrm{GC}} = 2^{-28}$. They are dyadic scales derived from the training residual and readout-Jacobian singular values, using only the same fourteen distributed-viscosity trajectories. The momentum-weak balance is centered about the actual scalar H_4 before adding the network perturbation. The perturbation is evaluated directly, rather than obtained by subtracting full and baseline floating-point forces. The baseline still contains the arithmetic of the native observations and quadrature. Its quadratic jet is evaluated from the actual scalar H_4, rather than regrouped through the stored approximation to K_2. The sum of the normalized weak loss and the 0.1-weighted kinematic loss is divided by its fixed H_4 value. There is no raw-parameter penalty. The matrix head remains fixed, and the kinematic and damping terms retain the actual full-phase $H_{s,p}$.

Both fits complete 400 full-batch Adam updates with learning rate 0.01 and $\epsilon = 10^{-8}$. Every layer remains trainable. The initial zero readout gives zero hidden-layer data gradients at the first update; nonzero hidden-parameter and prediction changes are recorded after the second update. Checkpoint selection again minimizes the training total over the initial state and all recorded updates. The selected updates are 400 for GR and 385 for GC, with training totals

Table 5. Generic-network gradient-vector RMSE on the same forty reused rest-point labels. Entries are $10^3 E_s$, in common dimensionless potential-gradient units. The final row is the arithmetic seed mean.

Seed	R	C	GR	GC
43	7.219805	6.034963	0.099531	0.009317
47	9.520357	6.648840	0.298038	0.007967
53	12.884718	5.769706	1.104723	0.010986
Mean	9.874960	6.151169	0.500764	0.009423

Table 6. Five paired contrasts of the RMSE in Table 5, multiplied by 10^3. For $A - B$, a positive value means that B has smaller error. The interaction is $(\mathrm{GR} - \mathrm{GC}) - (\mathrm{R} - \mathrm{C})$. Means use all three paired seeds.

Contrast	Seed 43	Seed 47	Seed 53	Mean
R−GR	7.120274	9.222319	11.779995	9.374196
C−GC	6.025646	6.640873	5.758720	6.141746
R−C	1.184843	2.871517	7.115013	3.723791
GR−GC	0.090214	0.290071	1.093737	0.491341
Interaction	−1.094628	−2.581446	−6.021276	−3.232450

Table 7. Reused full-phase diagnostics. Weak and kinematic columns are pooled normalized RMS values over twelve seed–trajectory combinations per cell. The rest-velocity range covers the three seedwise forty-point RMS values and is in the original dimensionless velocity coordinates.

Cell	Momentum weak	$H_{s,p} - v$	Rest $H_{s,p}(Q,0)$ RMS range
R	5.310319×10^{-2}	1.255422×10^{-1}	$[1.518317 \times 10^{-3},\ 5.222680 \times 10^{-3}]$
C	3.948955×10^{-2}	6.424029×10^{-2}	$[1.484900 \times 10^{-3},\ 2.417393 \times 10^{-3}]$
GR	4.187033×10^{-3}	1.810586×10^{-2}	$[1.498658 \times 10^{-4},\ 3.086894 \times 10^{-3}]$
GC	5.695685×10^{-5}	3.301787×10^{-4}	$[8.301210 \times 10^{-6},\ 2.276685 \times 10^{-5}]$

relative to the initial value of approximately 4.1951×10^{-4} and 1.9017×10^{-2}, respectively.

Both selected outputs are frozen before their evaluation against the same forty previously used labels. No additional held-out dataset is introduced or used for selection in this comparison. It is exploratory reused validation, rather than a new blind test, and does not replace the twelve fits or five paired contrasts above. Only these forty labels are evaluated; no trajectory-holdout evaluation is performed for the two repaired outputs.

The real model has the smaller error in this comparison, and neither learned output improves on the shared prior (Table 8). Together with the original factorial result, this reversal demonstrates sensitivity to the stated training scales and initialization; it does not establish an intrinsic advantage of either representation. The repaired GC output additionally admits a separate finite configuration-slice and full-phase error analysis in Supplement Section H. Those bounds concern this new checkpoint and do not transfer the conservative KAM or sampled-controller conclusions for the structured model.

4 Conservative KAM certification of the frozen pair

Ordinary KAM theory applies here to the canonical Hamiltonian fields with input and dissipation set to zero. Define

$$H_\lambda = \widehat{H} + \lambda(H_* - \widehat{H}), \qquad 0 \le \lambda \le 1. \tag{34}$$

This family retains the same prescribed inertia and two fixed endpoint potentials. Classical nonresonance and action-frequency nondegeneracy are substantive requirements (Pöschel 2001); positive bending stiffness alone does not establish them.

We fix period-one angles and $\nu = \omega/(2\pi)$, where

$$\begin{aligned} \omega_1 &= \frac{499974648628101}{500000000000000}, \\ \frac{\omega_2}{\omega_1} &= [6; 6, 1, 1, 1, 2, 1, 5, 14, 17, 4, \overline{1}]. \end{aligned} \tag{35}$$

The bar denotes an infinite golden continued-fraction tail. This exact choice, made from the fourth-order normal form before Newton correction, has a verified all-integer Diophantine bound $|k \cdot \nu| \ge \gamma_D |k|_1^{-1}$ with $\gamma_D = 4828522443384859/5764607523034234 88$. The small-divisor constant is unrelated to the viscosity coefficient.

For the frozen learned torus K_0, the accepted cubic response is

$$\begin{aligned} K_\lambda^{[3]} &= K_0 + \lambda U_1 + \lambda^2 U_2 + \lambda^3 U_3, \\ E_\lambda &= X_{H_\lambda}(K_\lambda^{[3]}) - D K_\lambda^{[3]} \nu. \end{aligned} \tag{36}$$

Table 8. Precision-centered comparison on the same forty reused rest labels. Gradient-vector RMSE uses the normalization in (33). Neural derivatives are ordinary floating-point scalar-energy automatic derivatives; the table is not an interval ranking certificate.

Model	Selected update	Rest-gradient vector RMSE
GR	400	$1.427681587 \times 10^{-11}$
GC	385	$9.819100531 \times 10^{-11}$
Shared H_4	—	$1.297916885 \times 10^{-11}$

The residual norm is the supremum of the component magnitudes over the entire complex angular strip of width $\rho = 1/40$. Low Fourier interpolants, Fourier aliasing, analytic tails and the parameter tail are all included. A real grid or a small Hamiltonian error cannot replace this invariance-defect bound.

The observation bound (27) supplies configuration derivatives through order three on $|\operatorname{Re} q_i| \le 0.06$, $|\operatorname{Im} q_i| \le 0.01$. Because the kinetic parts coincide, these give the vector-field, Jacobian and ordered second-field-derivative discrepancies. The transported first-field bound is below 8.324054×10^{-12}. Adding these bounds to the learned-field bounds on the common embedding tube rechecks the Gram and mean-torsion inverse margins and all dependent constants in Haro and Luque (2019, Theorem 3.1). The parameter-tail modulus is recomputed on the retained complex λ disk. A separate, larger-domain analytic estimate supplies the Fourier tail; the smaller native error rectangle is not assumed to cover that domain.

Theorem 3. Certified conservative family. *For each real $\lambda \in [0,1]$, the specified finite-Gauss H_λ has a real analytic embedded invariant two-torus with the fixed frequency (35), analytic on the angular strip $\rho_\infty = 1/400$.*

The interval and rational calculation verifies the full canonical hypotheses and eight scalar gates of the cited a posteriori theorem. The recomposed complete residual is below 1.205897×10^{-41}, while the sufficient threshold exceeds 5.82604×10^{-38}. Supplementary equations (S18)–(S20) describe how the native learning bound enters the transport. The small residual belongs to the corrected embedding; it is distinct from the approximately 10^{-12} model discrepancy. The conclusion gives neither a phase-space percentage nor a torus for the positively damped or forced flow. It also does not assert analytic dependence of the exact tori on λ.

4.1 Fresh point observations for the same conservative family

The independent forty-configuration validation of Section 3.3.5 closes the point-observation identity (28) for the unchanged learned Hamiltonian. The controlled-reference label errors, the recovered signed principal and both analytic-tail contributions bound every complex partial through order three. The four ordered increments use the same tensor-norm factors as the weak-observation route. Recomputing the Gram and mean-torsion margins, parameter tail and eight canonical KAM inequalities gives a largest gate below 2.205303×10^{-4} for the same family, frequency and cubic response. Its complete residual remains below 1.205897×10^{-41} and retains the independent Fourier and large-domain analytic estimates.

This acceptance uses the realized finite point inverse and certified reference-fidelity allowances. It supports the same conservative-family conclusion without asserting a continuous-density or full-network statistical generalization theorem. The frozen-output population-risk calculation below supplies a separate function-level verification.

4.2 A verified frozen-output population-risk premise

The same Hamiltonian pair also admits direct integration of a population risk. With Q uniform on $[-1/16, 1/16]^2$, define

$$R_6 = \left(\mathbb{E}_Q \sum_{i=1}^{2} |\partial_i (\widehat{V} - V_*)(Q)|^6 \right)^{1/6}. \qquad (37)$$

Independently enclosed Taylor coefficients, exact normalized polynomial moments and analytic remainders prove $R_6 < 2.395279 \times 10^{-14}$. A polynomial L^2-to-complex-domain estimate then gives derivatives through order three, as proved in Supplementary Proposition S11. They satisfy the predeclared factor-512 sufficient derivative budget for the same pair. Recomputation of its mean torsion, parameter tail and all eight canonical KAM gates passes, with largest gate below 2.385535×10^{-4}. The complete embedding residual remains below 1.205897×10^{-41} and retains its independent Fourier and large-domain analytic premises.

This calculation verifies a frozen-output risk premise by access to the known reference function. It supplies a function-level route into the same KAM proof, without inferring the risk from empirical observations or instantiating a class-uniform statistical learning theorem.

4.3 The conditional statistical route

The statistical connection of Chen et al. (2022) is retained as a separate component in the supplementary material. On a bounded connected Lipschitz phase domain of dimension four, an iid gradient-loss risk bound, a positive density lower bound, and a hypothesis-class bound imply a gauge-fixed energy bound by first-order Sobolev embedding with integrability exponent $p > 4$. A Hessian bound then supplies an interior gradient bound; a holomorphic outer-strip bound and interpolation produce the analytic residual needed by the a posteriori theorem. The complete argument keeps domain loss, target-torus defect and smallness conditions explicit. The exponent $p > 4$ is not an assertion of fourth-order differentiability.

For the unrestricted continuous raw-parameter implementation, the real-state imaginary-row guards alone provide neither a bounded loss envelope nor a common prescribed complex domain. Supplementary Propositions S7 and S8 give explicit members demonstrating these two obstructions. The class-uniform statistical implication therefore requires additional parameter-class conditions beyond a certificate for one frozen output.

The present prescribed trajectory windows do not instantiate that iid state-sampling law or a full-dimensional density bound. The established learning-to-KAM connection is therefore the deterministic fixed-design route above. The known-reference risk calculation verifies a separate frozen-output premise. The conditional statistical theorem explains the additional premises needed for a probability statement; it is not used to assign one to these data.

5 Power transport and sampled magnetic feedback

The dissipative structure has its own conclusion. Along the learned equations, canonical cancellation and (8) give

$$\widehat{H}' = -v^{\mathsf{T}}\widehat{R}v + uy, \qquad y = g(q)^{\mathsf{T}}v. \tag{38}$$

For the physical reference under the same input, let $|\partial_i(\widehat{V} - V_*)| \le \varepsilon_i$ and $\left\|\widehat{M} - M_*\right\|_2 \le \delta_M$, with $M_* = S^{-1}R_*S^{-\mathsf{T}}$. Then

$$\widehat{H}' \le -v^{\mathsf{T}}\widehat{R}v + uy + \sum_i \varepsilon_i|v_i| + \delta_M v^{\mathsf{T}}Gv. \tag{39}$$

The last term follows by setting $z = S^{\mathsf{T}}v$. If $M_* \succeq m_*I$ and $\delta_M < m_*$, then $\widehat{M} \succeq (m_* - \delta_M)I$ and $v^{\mathsf{T}}R_*v \ge [m_*/(m_* + \delta_M)]v^{\mathsf{T}}\widehat{R}v$. Thus identified accuracy quantifies physical dissipation credit, whereas PSD is already imposed by the decoder.

5.1 Continuous law and the distinct held-flow map

Write $s = d^{\mathsf{T}}q$, $s_* = 3/1000$ and $\mathrm{sat}_a(z) = \max(-a, \min(a, z))$. The frozen learned-equilibrium feedforward is $T = 6380576977437163/144115188075855872$. With $(k_p, k_d, a_p, a_d) = (4, 3, 1/50, 3/100)$, use

$$\pi(x) = \frac{T - \mathrm{sat}_{a_p}(k_p(s - s_*)) - \mathrm{sat}_{a_d}(k_d s')}{\cos s}. \tag{40}$$

The reference shaped storage $J_b = H_* - Ts + \int_0^{s-s_*} \mathrm{sat}_{a_p}(k_p r)\,\mathrm{d}r$ obeys $J_b' = -v^{\mathsf{T}}R_*v - s'\,\mathrm{sat}_{a_d}(k_d s') \le 0$. Strict potential convexity, a certified interior equilibrium and an energy barrier establish continuous-reference convergence on the stated compact invariant component. The declared initial state $q_0 = (1/80, 1/80)$, $p_0 = (1/200, 1/1000)$ lies in that component, where $|\pi| < 0.094416 < 0.1$.

Holding a command adds $(u_n - \pi(x))\cos s\, s'$ to the storage derivative. Consequently that continuous proof does not prove sampled stability. For the exact constant-input flow Φ_h, the ideal sample map is $\mathcal{S}_h(x) = \Phi_h(x, \pi(x))$. Its derivative uses the held-input plant matrix and the input sensitivity, rather than the exponential of the continuous closed-loop Jacobian. Complete held-flow and variational enclosures give a fixed dyadic P_L with $4I \prec P_L \preceq 74.237I$ and the local ellipsoid

$$\begin{aligned} V_L(x) &= (x - x_e)^{\mathsf{T}}P_L(x - x_e), \\ \mathcal{E} &= \{V_L \le 81 \times 10^{-6}\}. \end{aligned} \tag{41}$$

For $h \in [0.1 - 10^{-12}, 0.1 + 10^{-12}]$, $V_L(\mathcal{S}_h(x)) \le 0.987115V_L(x)$ on $\mathcal{E}$. An outer-box first-exit argument covers the entire hold. If the actual held command differs from $\pi(x_n)$ by at most $\epsilon_u = 10^{-5}$, the same analysis gives

$$\begin{aligned} r_{n+1} &\le c_s r_n + L_u\epsilon_u, \qquad r_n = \sqrt{V_L(x_n)}, \\ c_s &= 497/500, \qquad L_u = 224417/10^6. \end{aligned} \tag{42}$$

This proves local practical sample invariance. Initial entry from the experiment is verified separately, not inferred from the local theorem.

5.2 A continuing numerical state and energy ledger

Observation and numerical evaluation must establish the command-error premise at every sample. The continuing program admits finite binary64 centers z_n with $\|z_n - x_n\|_\infty \le 10^{-7}$. Exact rational metric-polynomial and cosine enclosures yield a nominal request on the 2^{-40} grid. Uniform bounds on actual interval returns prove evaluation error below 4.547474×10^{-13}. Including observation sensitivity, the command discrepancy is below $8.804106 \times 10^{-7} < \epsilon_u$. These are uniform interface bounds on an explicit domain, not repeated point checks.

The ledger uses grid $g_E = 2^{-80}$ and runtime cap $c_{\mathrm{rt}} \simeq 0.002370501997$. At a sample it takes the minimum of the carried physical-energy bound and a fresh outward learned-energy candidate, including the frozen conservative-error allowance $\varepsilon_0^{\mathrm{run}}$. Both operands bound the same physical energy, so this reset is valid. With $\widetilde{y}$, e_y and $\underline{d}$ enclosing the port value, its error and physical dissipation, respectively, the scalar work gate requires

$$\begin{aligned} h_+(u\widetilde{y} + |u|e_y - \underline{d}) + \tfrac{1}{2}h_+^2(|u|L_y + L_d) \\ \le \tfrac{1}{2}(c_{\mathrm{rt}} - W_n^{\mathrm{reset}}). \end{aligned} \tag{43}$$

Here L_y, L_d are physical derivative bounds and $h_+ = 0.1 + 10^{-12}$. Upward rounding and a maximum with the reset ledger bound the entire hold, including durations shorter than h_+. The executed algorithms retain their frozen full-box identification budgets. Later local error bounds are not silently substituted into a past decision.

Theorem 4. Conditional continuing practical invariance. *Suppose $x_0 \in \mathcal{E}$, $H_*(x_0) \le W_0 < c_{\mathrm{rt}}$ and $W_0 \in g_E\mathbb{Z}$. At every sample assume the stated observation inclusion, exact constant execution of the returned command, a hold in the stated clock window, the fixed physical reference and the supplementary computational premises. The continuing program returns normally and maintains $x_n \in \mathcal{E}$ and $H_*(x_n) \le W_n < c_{\mathrm{rt}}$. On every entire hold, $H_* < 0.001441122076$ and $|q_i| < 69/1250$.*

The proof combines the uniform numerical interfaces, the work inequality, the recurrence (42), and energy/configuration first-exit barriers in a joint induction. It does not assume the smaller learning domain to justify the ledger that establishes that domain. Supplementary Theorem S4 provides the full argument and precision/serialization assumptions.

A nonnegative comparison exponential also transports the sampled recurrence through each hold. Its ultimate generalized-angle bound is $\limsup |d^{\mathsf{T}}q - s_*| < 0.000107094841601$. Starting at index zero with any admissible initialization, index 306 is a sufficient bound for all subsequent entire-hold errors to be below $1/2000$. This is a conditional theoretical index, not an executed duration or a first hitting time. Persistent bounded errors give a practical bound rather than exact convergence.

5.3 Validated finite feedback and its continuing-control scope

The first feedback program executes 200 holds, independently validated over 20,000 integration slabs. A separate continuous query on its final fifty holds bounds the generalized-angle error $|d^T q - 3/1000|$ by $3.476316 \times 10^{-4} < 1/2000$ throughout those holds. This output is a generalized angle. The first guaranteed whole-tube entry into the local ellipsoid occurs at sample 199, leaving one actually executed post-entry hold in that record.

The separate continuing decision program subsequently executes two further COMSOL holds, computing each command from the preceding native endpoint. The issued values are

$$u_{200} = \frac{49692742889}{2^{40}}, \qquad u_{201} = \frac{49763151619}{2^{40}}. \tag{44}$$

Both command readbacks and complete native-state restarts agree bitwise. Independent reference-IVP verification covers all 200 additional slabs, carrying the positive error radius from the first record through the switch. At sample 202, the entire terminal tube satisfies $V_L \le 7.540 \times 10^{-5} < 81 \times 10^{-6}$ and retains a validated energy ledger. No state restart or energy reset removes the propagated IVP error.

Thus the executed evidence contains 200 holds under the preceding program and two under the continuing program. The latter's uniform numerical contracts support subsequent control under the stated observation, held-actuation and clock assumptions. Its entire-hold threshold guarantee from index 306 is conditional and is separate from the completed final-fifty-hold query. Persistent bounded errors yield practical bounds. These results address the specified numerical reference; statistical generalization, material calibration and a general complex-over-real advantage are not established by these experiments.

6 Discussion and conclusion

The framework keeps the learned functions, physical reference and mathematical endpoints aligned. The same frozen conservative potential appears in weak-observation inversion, complex-domain derivative transport and the conservative KAM family. The matrix head has an algebraic power sign and a separately quantified error against distributed viscosity. These properties enter physical feedback through explicit dissipation credit and error budgets. Sampled practical stability additionally requires the held-flow, numerical-interface and ledger arguments.

Several restrictions determine the interpretation. Inertia and the low-order potential jet are prescribed, and learning is sequential. The analytic reference is a finite quadrature reduction with an assumed viscosity, so numerical certification does not establish material calibration or convergence to a continuum rod. The Rayleigh decoder depends on configuration and is quadratic in velocity. More general velocity-dependent drag, contact and unknown external forces require different structural and robustness arguments. Offline COMSOL decision-and-integration experiments also do not demonstrate real-time hardware execution.

The real dissipation head performs better on the reported matrix identification metrics. The generic conservative factorial study gives lower rest-gradient errors for the complex models under both prior conditions in all three paired initializations. These comparisons concern different learning tasks. The generic architectures, pole constraints and initial function scales differ, and their learned outputs remain less accurate than the zero-correction fourth-order prior on this rest test. A separate precision-centered, common-H_4 initialization reverses the prior-assisted ranking: the real model has the smaller error, and both repaired outputs remain worse than H_4 on the reused observations. The generic comparison is therefore sensitive to the objective and initial function scale. Its new full-phase derivative and power bounds are stated separately in the supplement and do not replace the structured model's KAM or executed-feedback certificates. The certified structured potential remains the deployed energy. The fourteen-direction theorem supplies a positive sufficient neighborhood for finite-design observability. Point-gradient inversion provides a separate finite-output acceptance argument with explicit label, signed principal and analytic-tail terms. Its realized forty-configuration simulation validation satisfies these budgets and the recomputed KAM conditions for the unchanged frozen pair. The accompanying correction ablation shows a substantial improvement over the shared fourth-order prior on the same observations. The very short observation window and field-defined velocity observer make this a numerical validation; physical sensing would require its own error and clock contracts. The simultaneous acceptance theorem also permits candidates chosen using the same observations, with their own certified remainders and model-dependent downstream witnesses. Direct known-reference integration also verifies a population-risk premise for the frozen pair and transports it through the complete retained KAM gates. This oracle calculation does not establish that sampled empirical risk controls population risk, and none of these statements guarantees a successful arbitrary optimization run. The conditional statistical theorem retains its additional sampling, class-complexity and regularity premises.

Within these limits, native observations support deterministic error bounds for a trained complex mechanical model, conservative invariant tori for its frozen reference family, and separate dissipative power and sampled magnetic-feedback conclusions. Their common mechanical definitions make the proof obligations and experimental scope explicit.

Data and code availability

The training checkpoints, native COMSOL exports, exact-rational and interval certificates, and verification code are indexed by the SHA-256 reproducibility manifest accompanying the manuscript. The supplementary material specifies the mathematical assumptions, full proofs and numerical-certificate conventions.

Declarations

This study reports mathematical analysis and simulated mechanical data. It involved no human participants or animals.

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